

The Art of Solving Thermodynamics Problems

RP's Method — a disciplined way to plan your attack

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RP's method This note sets out *one* way to plan a thermodynamics problem — the way I teach it. It is not the only correct way, but shares its foundations with the worked examples you will see elsewhere in the unit: the same first law, the same sign convention (heat in and work out positive), the same distinction between *state properties* (P, T, v, u) and *process variables* (q, w), and the same discipline of stating assumptions before computing. What this method adds is a single organiser — the **arrow diagram** — that makes the plan visible on the page before any number is worked.

1. Why bother

Engineering design and innovation are, at heart, the solving of complex puzzles. A sudoku is a small, familiar example: you are handed a few numbers and a short list of rules, and from those you deduce, square by square, everything left blank. A thermodynamics problem is the same. You are given a few properties, you know a fixed set of physical rules, and your task is to work out everything else. Fundamentally, sudok puzzles, or geometry theorems, or thermodynamics problems are all exercises in deductive logic.

Every puzzle of this kind gives way to the same three moves:

1. learn the rules of the game;
2. take stock of the information you have actually been given;
3. combine the rules, that information, and a little logic to deduce something new — then repeat, each new fact opening the next, until the answer falls out.

Thermodynamics is an unusually good training ground for this. Its rules are not trivial: they reward real understanding and disciplined, systematic use rather than guesswork. Build the habit of solving thermodynamics problems methodically and the same habit carries into almost any engineering puzzle you meet later. That is why it is worth treating the subject as a *dojo*.

The one habit this whole method is built around. In a thermodynamics problem most of the work — perhaps eighty percent of it — is the *planning*, and that planning happens on your rough worksheet before you compute a single number. Once the plan is laid out, putting numbers into the equations is trivial bookkeeping. Skip the planning and reach straight for the numbers, and you tend to get lost halfway through: your clean answer sheet quietly turns into a second rough worksheet. So separate the two jobs. Think and plan on the rough sheet first; then transcribe a clean solution.

The rest of this note is that plan, made explicit. We stay throughout with the **Perfect Ideal Gas** (PIG).

2. The rules of the game: the five buckets

Every closed-system calculation for an ideal gas is built from five kinds of equation. Keep these five *buckets* in mind: everything you write comes from one of them, and the source-tag you jot in the margin against each equation *is* its bucket.

1. **Definitions** — e.g. enthalpy $h = u + Pv$, and the specific heats.

2. **Equations of state** — for an ideal gas, $Pv = RT$ and $u = c_v T$. (An entropy EoS exists too, but for the PIG cycles in this unit we never need to compute s , so we leave it aside.)
3. **Conservation of energy** — the first law, $\Delta u = q - w_b$ for a closed system.
4. **Work and heat relations** — boundary work $w_b = \int P dv$, evaluated on the actual process path.
5. **Process specifications** — constant volume, constant pressure, constant temperature, or isentropic (a quasi-static adiabatic).

Solving a problem is then a puzzle in these rules. You know the equations in the buckets; you identify what is given and what is unknown; you use the rules to deduce the unknowns one step at a time. The game is to keep assembling equations until you have as many as you have unknowns.

3. The method, step by step

We will state each step in general, then carry a single example through every one of them. The example is the one worked live in the video:

Worked example. Air in a pneumatic cylinder starts at 100 kPa and 300 K. It is first compressed *isothermally* to a specific volume of $0.2 \text{ m}^3/\text{kg}$. The piston is then locked and the cylinder is cooled at *constant volume* until the temperature falls to 250 K. Per kilogram, find the net work output and the net heat input.

Step 0 — which ideal gas?

Every single time: identify the working fluid and which ideal gas it is, then write down its constants from Table A-2 — the gas constant R , the specific heat c_v , then $c_p = c_v + R$ and $k = c_p/c_v$.

For our example: the fluid is air, so $R = 0.287$, $c_v = 0.718$, $c_p = 1.005 \text{ kJ}/(\text{kg K})$ and $k = 1.4$. The words “pneumatic cylinder” also tell us this is a **closed system**, so $\Delta u = q - w_b$ applies.

Step 1 — states and processes

Find the key states, number them, and count them. Then identify the canonical process joining each pair. Every process you meet is one of four kinds: constant volume, constant pressure, constant temperature, or isentropic.

For our example: there are **three states** and **two processes**. State 1 at 100 kPa, 300 K. **1→2 isothermal**. State 2 at $v = 0.2 \text{ m}^3/\text{kg}$. **2→3 constant volume**. State 3 at 250 K. The path does not return to State 1, so there is no cycle and no efficiency to compute.

Step 2 — the arrow diagram

Lay the states out left to right. Above each state write the P , v , T you know and a question mark for the ones you do not; and record *where each property will come from* — given, from the **process**, or from the **equation of state**. Under each arrow write the process and its $P(v)$, with any heat or work you are told. This diagram is the plan.

State 1	State 2	State 3
$P_1 = 100$ (given)	$v_2 = 0.2$ (given)	$T_3 = 250$ (given)
$T_1 = 300$ (given)	$T_2 = T_1$ (process)	$v_3 = v_2$ (process)
$v_1 = RT_1/P_1$ (EoS)	$P_2 = RT_2/v_2$ (EoS)	$P_3 = RT_3/v_3$ (EoS)



isothermal

$$\begin{aligned}\Delta u &= 0 \\ w &= RT_1 \ln(v_2/v_1) \\ q &= \Delta u + w\end{aligned}$$

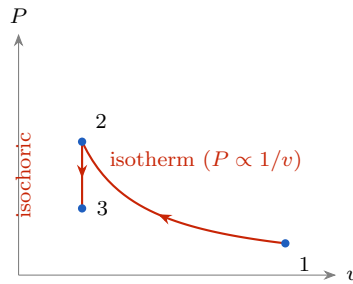
isochoric

$$\begin{aligned}\Delta u &= c_v(T_3 - T_2) \\ w &= 0 \\ q &= \Delta u\end{aligned}$$

Read the middle column: v_2 is given, T_2 is handed over by the **process**, and the **equation of state** closes it. That is the pattern the whole method is built to make automatic.

Step 3 — the P–v sketch

Sketch the path on a pressure–volume diagram. The arrow diagram fixed the *topology* (the order of states); the P–v sketch shows the *shape* of each path and checks that the processes join up sensibly. Remember the boundary work of each process is the area under its curve.



Stage 1 — fix P, v, T at every state

Now the calculation, in two stages. Stage 1 fixes P , v , T at every state. Start at the state where you already know two of the three; the ideal-gas EoS gives the third. At each next state, the given information supplies one property, the process supplies a second, and the EoS gives the third. Do not put in numbers yet — just write the equation under each state (that is what the arrow diagram already did).

For our example, carrying the numbers through:

$$v_1 = \frac{RT_1}{P_1} = \frac{0.287 \times 300}{100} = 0.861 \text{ m}^3/\text{kg}, \quad P_2 = \frac{RT_2}{v_2} = \frac{0.287 \times 300}{0.2} = 430.5 \text{ kPa},$$

$$P_3 = \frac{RT_3}{v_3} = \frac{0.287 \times 250}{0.2} = 358.8 \text{ kPa}.$$

Stage 2 — Δu , work and heat for every process

With P , v , T known, the energies follow. For every process, three equations: the internal-energy change $\Delta u = c_v \Delta T$; the boundary work $w_b = \int P dv$ along that path; and the heat $q = \Delta u + w_b$. The only equation that differs between processes is the boundary work:

constant volume	$w_b = 0$
constant pressure	$w_b = P \Delta v$
isothermal	$w_b = RT_1 \ln(v_2/v_1)$
isentropic	$w_b = c_v(T_1 - T_2)$

For our example:

$$w_{12} = RT_1 \ln \frac{v_2}{v_1} = 0.287 \times 300 \times \ln \frac{0.2}{0.861} = -125.7 \text{ kJ/kg} = q_{12} \quad (\Delta u_{12} = 0),$$

$$w_{23} = 0, \quad q_{23} = \Delta u_{23} = c_v(T_3 - T_2) = 0.718 \times (250 - 300) = -35.9 \text{ kJ/kg}.$$

Finish

Add the legs together, keeping their signs. The net work is the sum of the boundary works; the net heat is the sum of the heats. From there you can get anything else the problem asks.

$$w_{\text{net}} = -125.7 + 0 = \boxed{-125.7 \text{ kJ/kg}}, \quad q_{\text{net}} = -125.7 - 35.9 = \boxed{-161.6 \text{ kJ/kg}}.$$

Both negative: work is done *on* the gas, heat is rejected *from* it. A good habit is one self-check across the whole path: $q_{\text{net}} - w_{\text{net}} = \Delta u_{13} = c_v(T_3 - T_1) = 0.718 \times (-50) = -35.9$, and indeed $-161.6 - (-125.7) = -35.9$. ✓

4. The plan in one line

Identify the ideal gas → find the states and processes → draw the arrow diagram → sketch the P–v diagram → fix P , v , T at every state → then Δu , work and heat for every process → finish.

The same five buckets and the same steps handle any Perfect-Ideal-Gas problem, whether it is an engine, a pneumatic system, or a full cycle. The planning does the hard part; the arithmetic that follows is just bookkeeping.